SRN=PES2UG24CS258

Lab 7 - Fuzzing for Input Validation Bugs

Heading 1: Hypothesis Test Failures (Edge Cases Discovered)

```
============================= FAILURES ==============================
_____________________ test_parse_int_list_fuzz ______________________

    @given(st.text())
>   def test_parse_int_list_fuzz(s):
                    ^^^

test_processor.py:5:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
test_processor.py:7: in test_parse_int_list_fuzz
    result = parse_int_list(s)
             ^^^^^^^^^^^^^^^^^
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

s = ''

    def parse_int_list(s):
>       return [int(x) for x in s.split(',')]
                ^^^^^^

E       ValueError: invalid literal for int() with base 10: ''
E       Failing test case: test_parse_int_list_fuzz(
E           s='',
E       )

processor.py:2: ValueError
_____________________ test_process_user_data_fuzz ____________________
```

Heading 2: Test Success After Code Hardening / Fixing

```
C:\Users\USER\Downloads\L6_L7_12_cloud_storage_tier-main\L6_L7_12_clo
ud_storage_tier-main\Lab7>C:\Users\USER\AppData\Local\Python\pythonco
re-3.14-64\python.exe -m pytest test_processor.py
======================== test session starts ========================
platform win32 -- Python 3.14.8, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\USER\Downloads\L6_L7_12_cloud_storage_tier-main\L6_
L7_12_cloud_storage_tier-main\Lab7
plugins: anyio-4.15.1, hypothesis-6.170.15
collected 3 items

test_processor.py ...                                              [100%]

========================= 3 passed in 1.23s =========================

C:\Users\USER\Downloads\L6_L7_12_cloud_storage_tier-main\L6_L7_12_clo
ud_storage_tier-main\Lab7>
```

```
test_processor.py:15:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
test_processor.py:16: in test_process_user_data_fuzz
    process_user_data(data)
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

data = {}

    def process_user_data(data):
>       return data.get("username").strip().lower()
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'strip'
E       Failing test case: test_process_user_data_fuzz(
E           data={},
E       )
```